

Lecture 10 - Sep 30

Review

Use combinatorial Proof to prove that for all $n \geq 1$,

$$\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$$
$$\sum_{k=1}^n \binom{n}{k} \binom{k}{1} = \binom{n}{1} 2^{n-1}$$

$\sum_{k=1}^n \binom{n}{k} \binom{k}{1}$: pick a group of k people, and pick one of the k people to be a leader

$\binom{n}{1} 2^{n-1}$: pick one person to be a leader, then pick $k - 1$ other people to be members only.

Rigorous Proof

Proof.

We have n students. We want to count # of study-groups such that each group has a chosen leader. On one hand, we have $\binom{n}{1}$ ways to select a leader and 2^{n-1} ways to select remaining group members.

Alternatively, # ways to select k -member study-group with a chosen leader is $\binom{n}{k} \binom{k}{1}$, $k \geq 1, \dots, n$. By the addition principle, we have $\sum_{k=1}^n \binom{n}{k} \binom{k}{1}$ ways in total.

Therefore,

$$\binom{n}{1} 2^{n-1} = \# \text{ study-groups with a chosen leader} = \sum_{k=1}^n \binom{n}{k} \binom{k}{1}$$

□

Vandermon't's identity

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

Proof.

We have a group of m men and n women. We want to select a group of r people.

On one hand, we can select r people from the total $m + n$ people in $\binom{m+n}{r}$ ways.

Alternatively, we can first select k men and $r - k$ women in $\binom{m}{k} \binom{n}{r-k}$ ways, for all $0 \leq k \leq m$. By the addition principle, $\sum_{k=0}^m \binom{m}{k} \binom{n}{r-k}$.

Therefore,

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{r-k} = \# \text{ ways to select } r \text{ people} = \binom{m+n}{r}$$

□

Extension of Pascal's identity

$$\frac{P(n, k)}{k!} = \binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\dots(n-k+1)}{k!}$$

For any real # α , integer $k \geq 0$

$$\binom{\alpha}{k} = \frac{(-1)(-2)\dots(-k)}{k!} = \frac{(-1)^k}{k!}$$

$$\begin{aligned} \binom{\frac{1}{2}}{k} &= \frac{\frac{1}{2} * -\frac{1}{2} * -\frac{3}{2}}{k!} \\ &= \frac{1 * -1 * -3 * -5 * \dots * -(2k-3)}{2^k * k!} \\ &= \boxed{\binom{\frac{1}{2}}{k} = (-1)^{k-1} * \frac{(2k-3)!!}{2^k * k!}} \end{aligned}$$

$$\binom{\alpha}{k} = \begin{cases} \frac{\alpha * (\alpha-1) * (\alpha-2) * \dots * (\alpha-k+1)}{k!} & k \geq 1 \\ 1 & k = 0 \\ 0 & k < 0 \end{cases}$$

Extension of pascal's identity

$$\binom{r}{k} = \binom{r-1}{k} + \binom{r-1}{k-1}$$

verify these algebreically

applications of pascal's formula

Two identities by repeatedly applying pascal's identity to itself (one for its first term, one for its second term)

Remark: $\binom{r}{r} = \binom{r}{r-k}$ does NOT apply for $r \in \mathbb{R}$.

TODO: prove why? (topic for next lecture)

Expanding Second term

$$\begin{aligned}
\binom{r}{k} &= \binom{r-1}{k} + \binom{r-1}{k-1} \\
&= \binom{r-1}{k} + \left(\binom{r-2}{k-1} + \binom{r-2}{k-2} \right) \\
&= \binom{r-1}{k} + \binom{r-2}{k-1} + \left(\binom{r-3}{k-2} + \binom{r-3}{k-3} \right) \\
&\quad \dots \\
&= \binom{r-1}{k} + \binom{r-2}{k-1} + \binom{r-3}{k-2} + \dots + \binom{r-k+1}{0}
\end{aligned}$$

Now replace r with $r + k + 1$

$$\begin{aligned}
\binom{r+k+1}{k} &= \binom{r+k}{k} + \binom{r+k-1}{k-1} + \dots + \binom{r}{0} \\
&\quad \boxed{\sum_{i=0}^k \binom{r+i}{i} = \binom{r+k+1}{k}}
\end{aligned}$$

Remark: The above formula applies for any $r \in \mathbb{R}$.

For $r \in \mathbb{Z}$ the following applies

$$\boxed{\sum_{i=0}^k \binom{r+i}{r} = \binom{r+k+1}{k}}$$

But NOT for $r \in \mathbb{R}$ since $\binom{r}{k} = \binom{r}{r-k}$ also does not work for $r \in \mathbb{R}$.

Expanding first term

$$\begin{aligned}
\binom{r}{k} &= \binom{r-1}{k} + \binom{r-1}{k-1} \\
&= \left(\binom{r-2}{k} + \binom{r-2}{k-1} \right) + \binom{r-1}{k-1} \\
&= \left(\binom{r-3}{k} + \binom{r-3}{k-1} \right) + \binom{r-2}{k-1} + \binom{r-1}{k-1} \\
&\quad \dots \\
\binom{r}{k} &= \binom{0}{k-1} + \binom{1}{k-1} + \dots + \binom{r-2}{k-1} + \binom{r-1}{k-1}
\end{aligned}$$

Now replace r with $n + 1$ and k with $k + 1$

$$\begin{aligned}
\binom{n+1}{k+1} &= \binom{0}{k} + \binom{1}{k} + \dots + \binom{n}{k} \\
&\quad \boxed{\sum_{i=0}^n \binom{i}{k} = \binom{n+1}{k+1}}
\end{aligned}$$